SANPC Policy Repository DMS

Technical Documentation

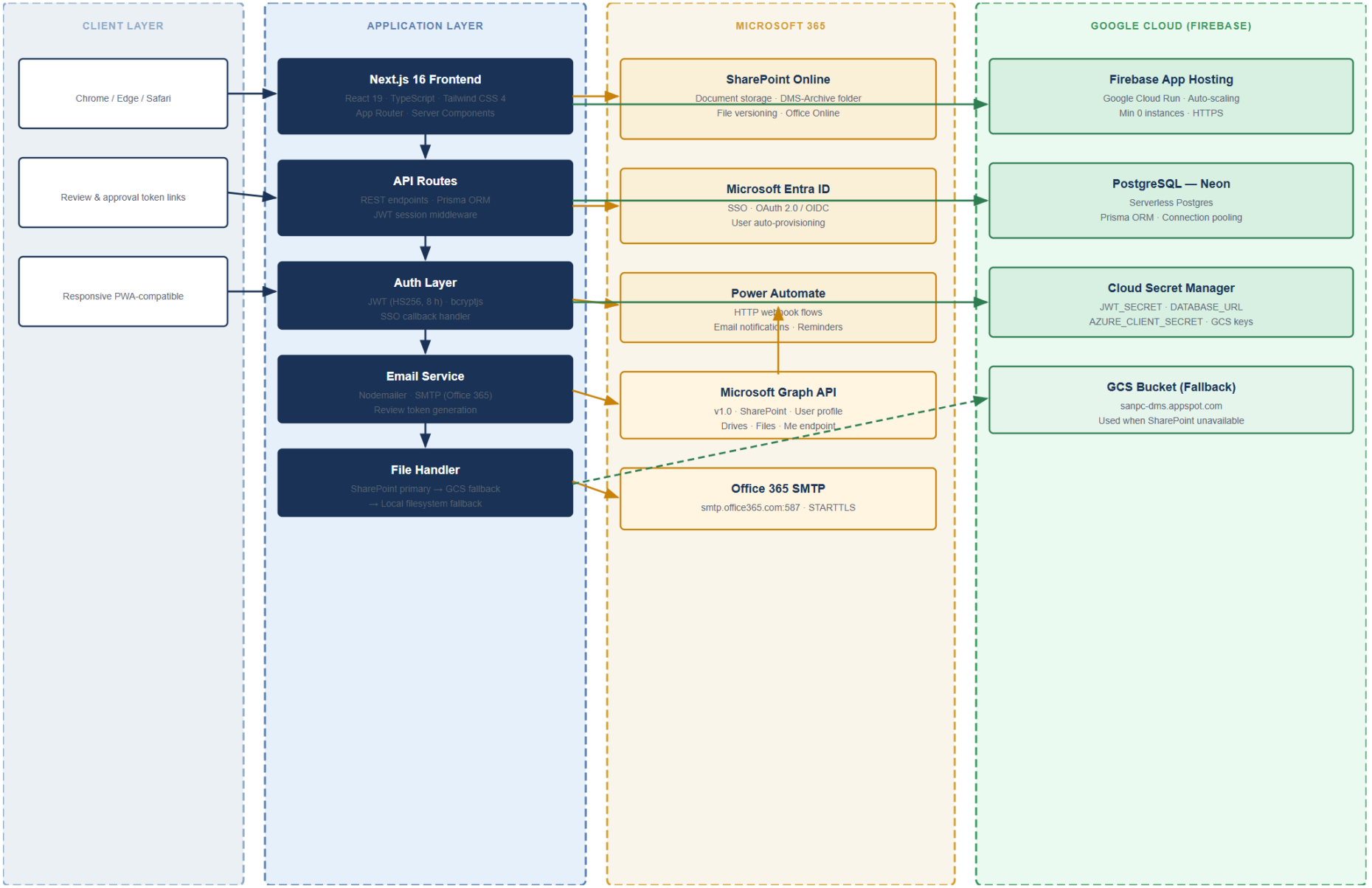
System Architecture • Use Cases • Sequence Diagrams • ERD • Status Flow

Organisation	South African National Petroleum Company (SANPC)
Document Ref	DMS-TECH-001 | Rev 00
Date	September 2026
Prepared By	ICT Department
Classification	Internal — Confidential

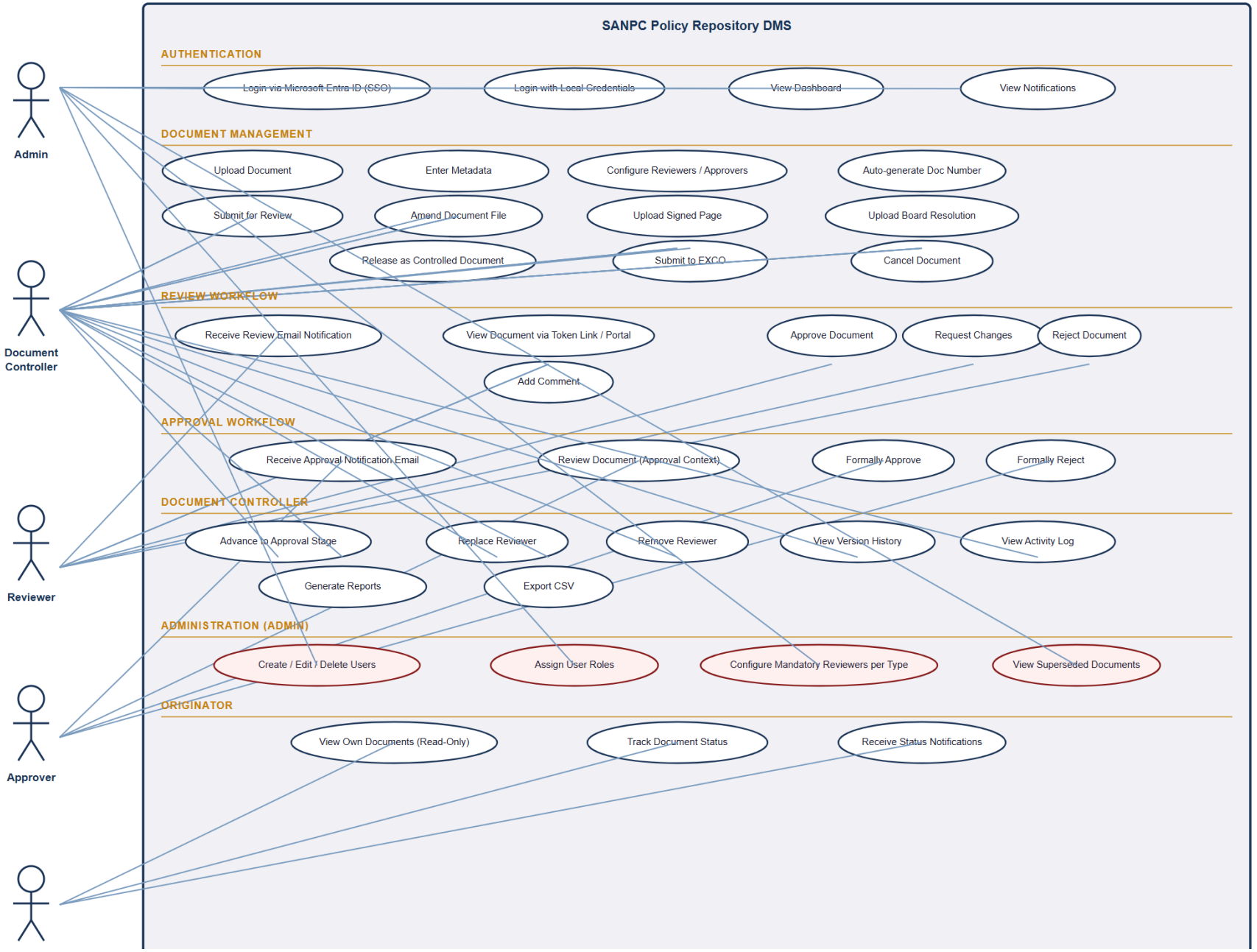
Table of Contents

1. System Architecture
2. Use Case Diagram
3. Sequence Diagram — Document Upload & Submission
4. Sequence Diagram — Review Decision Workflow
5. Sequence Diagram — Approval & Controlled Release
6. Sequence Diagram — Policy EXCO Workflow
7. Sequence Diagram — SSO Authentication Flow
8. Entity Relationship Diagram (ERD)
9. Document Status Flow

1. System Architecture

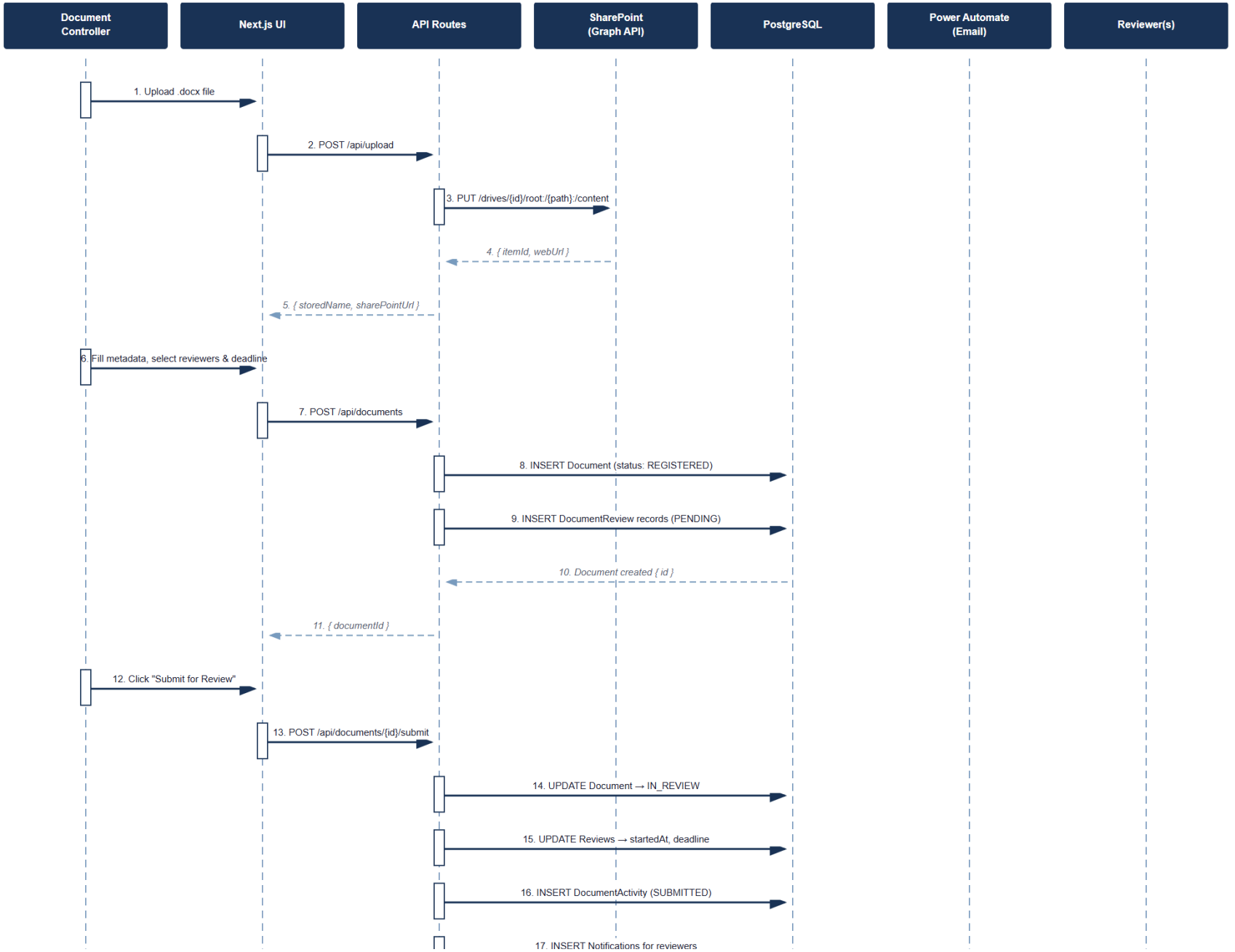


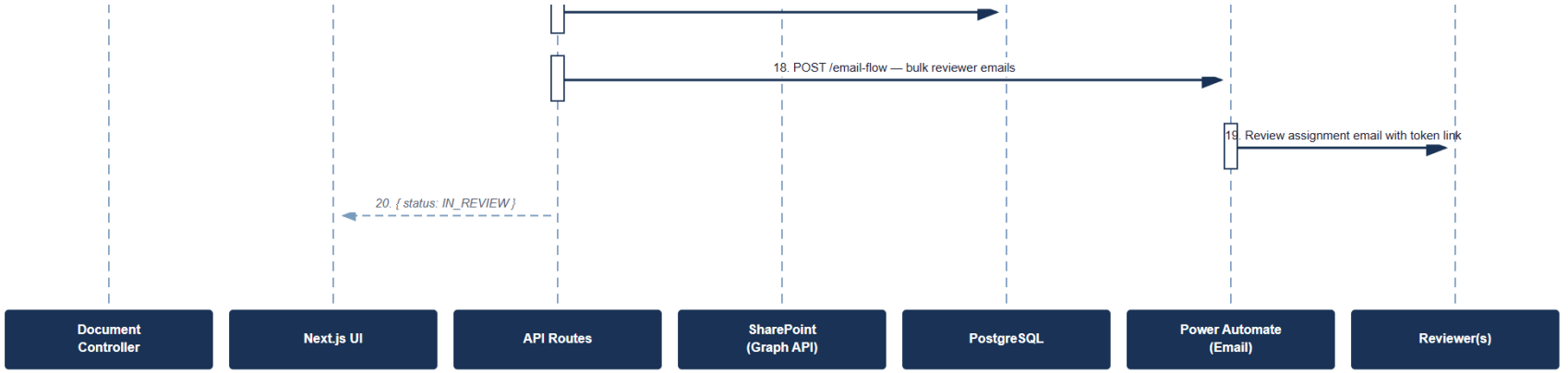
2. Use Case Diagram



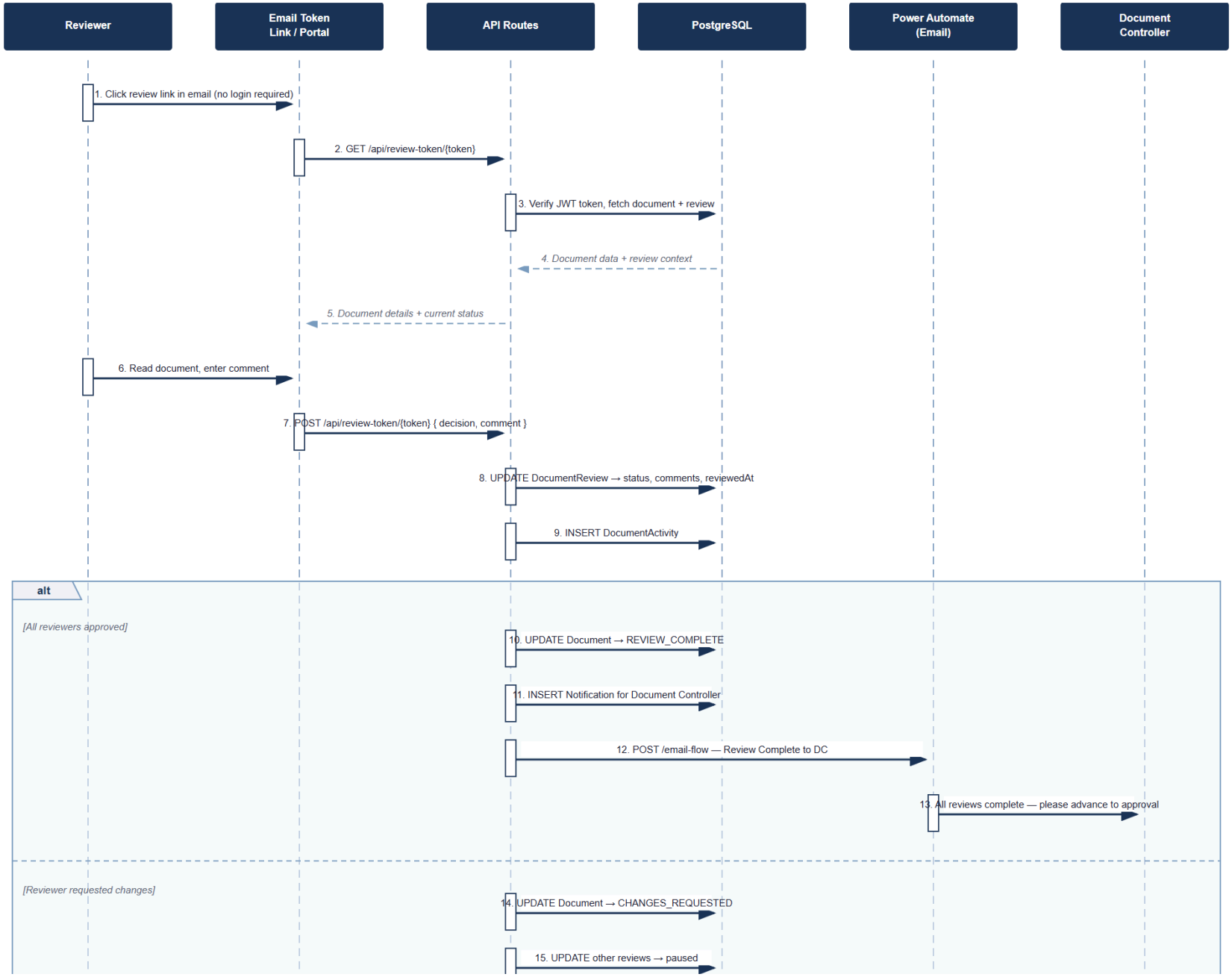

Originator
All Roles / Common
Admin Only

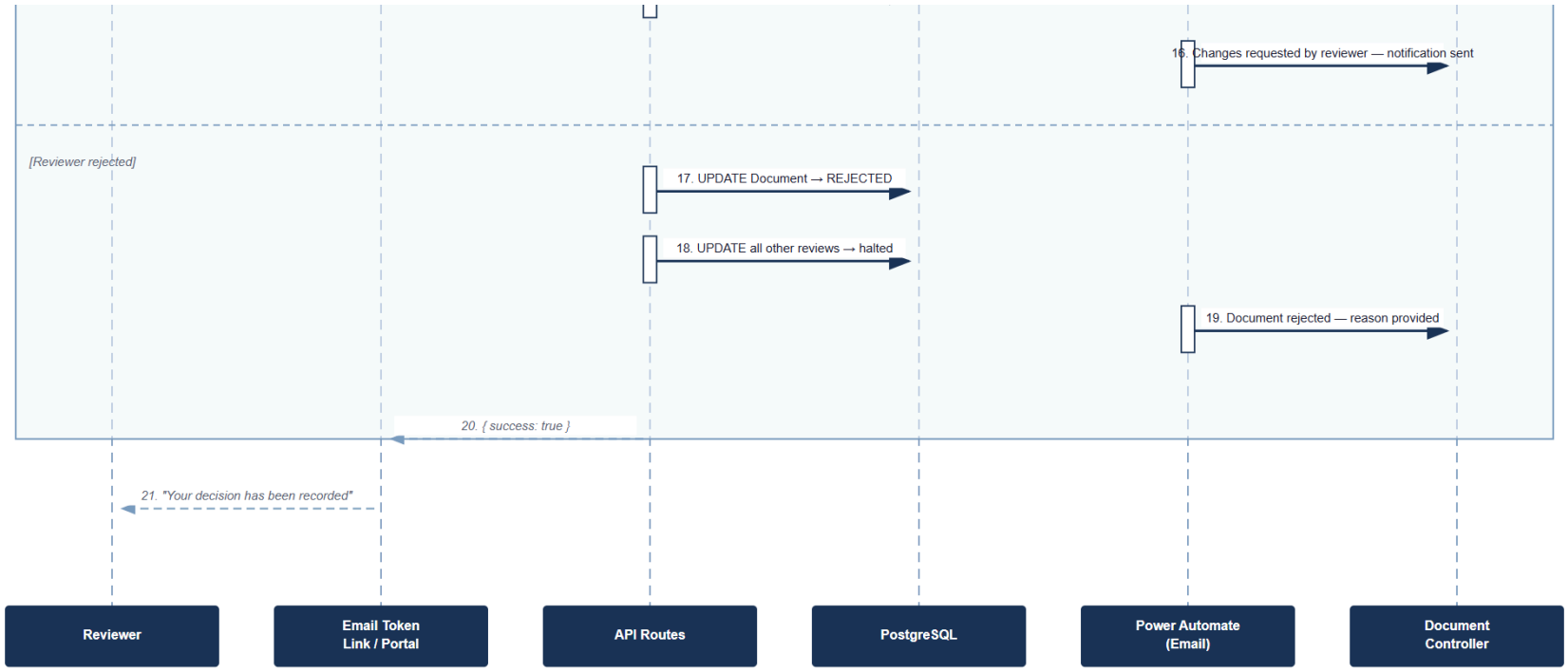
3. Sequence: Document Upload & Submission



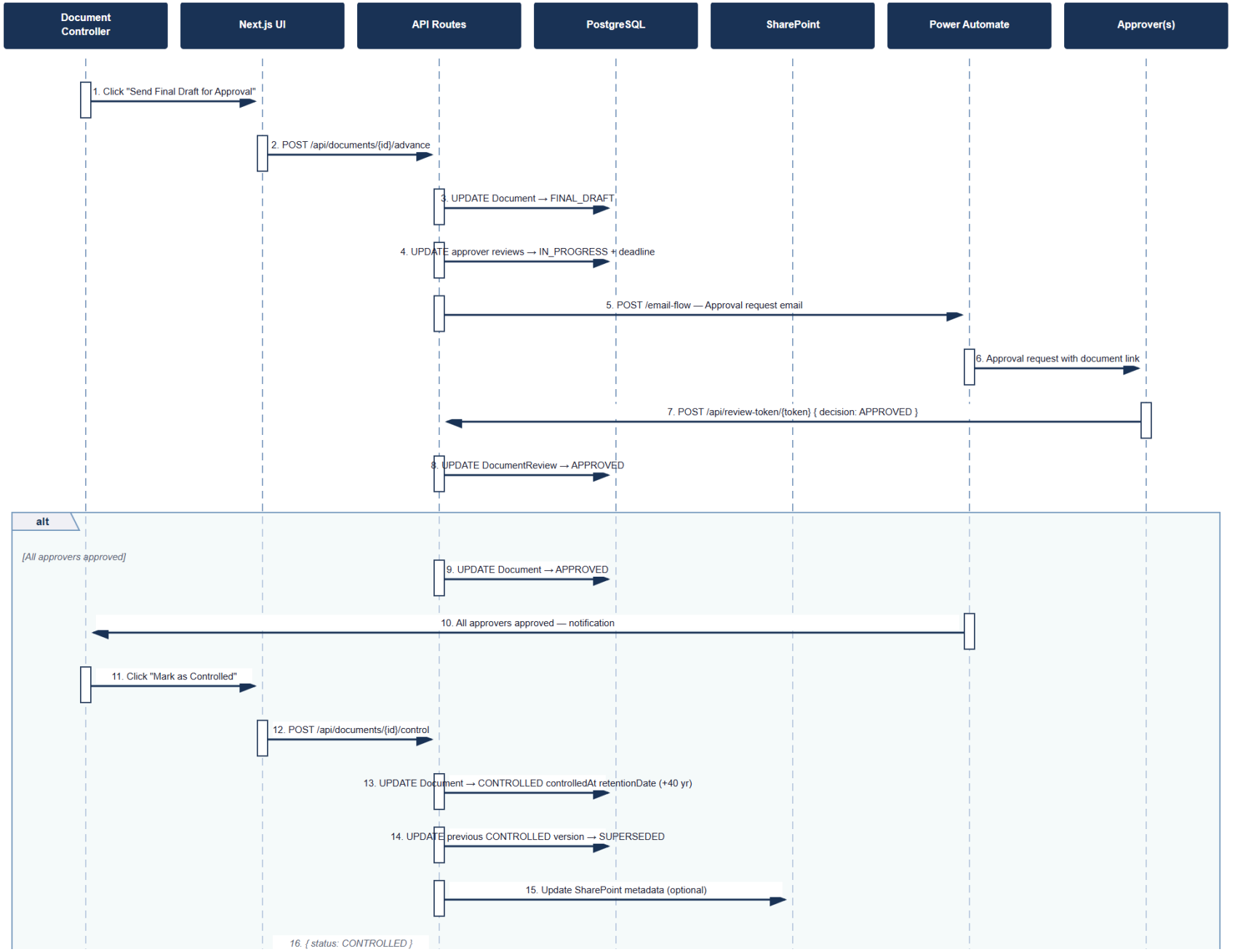

18. POST /email-flow — bulk reviewer emails
19. Review assignment email with token link
20. { status: IN_REVIEW }
Document Controller
Next.js UI
API Routes
SharePoint (Graph API)
PostgreSQL
Power Automate (Email)
Reviewer(s)

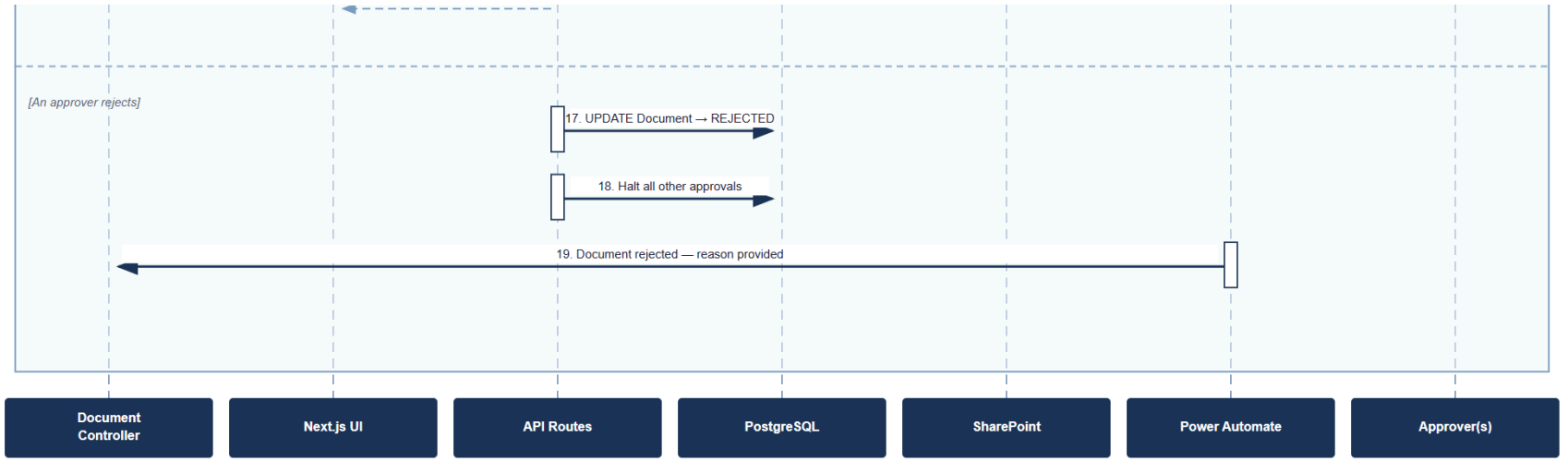
4. Sequence: Review Decision Workflow



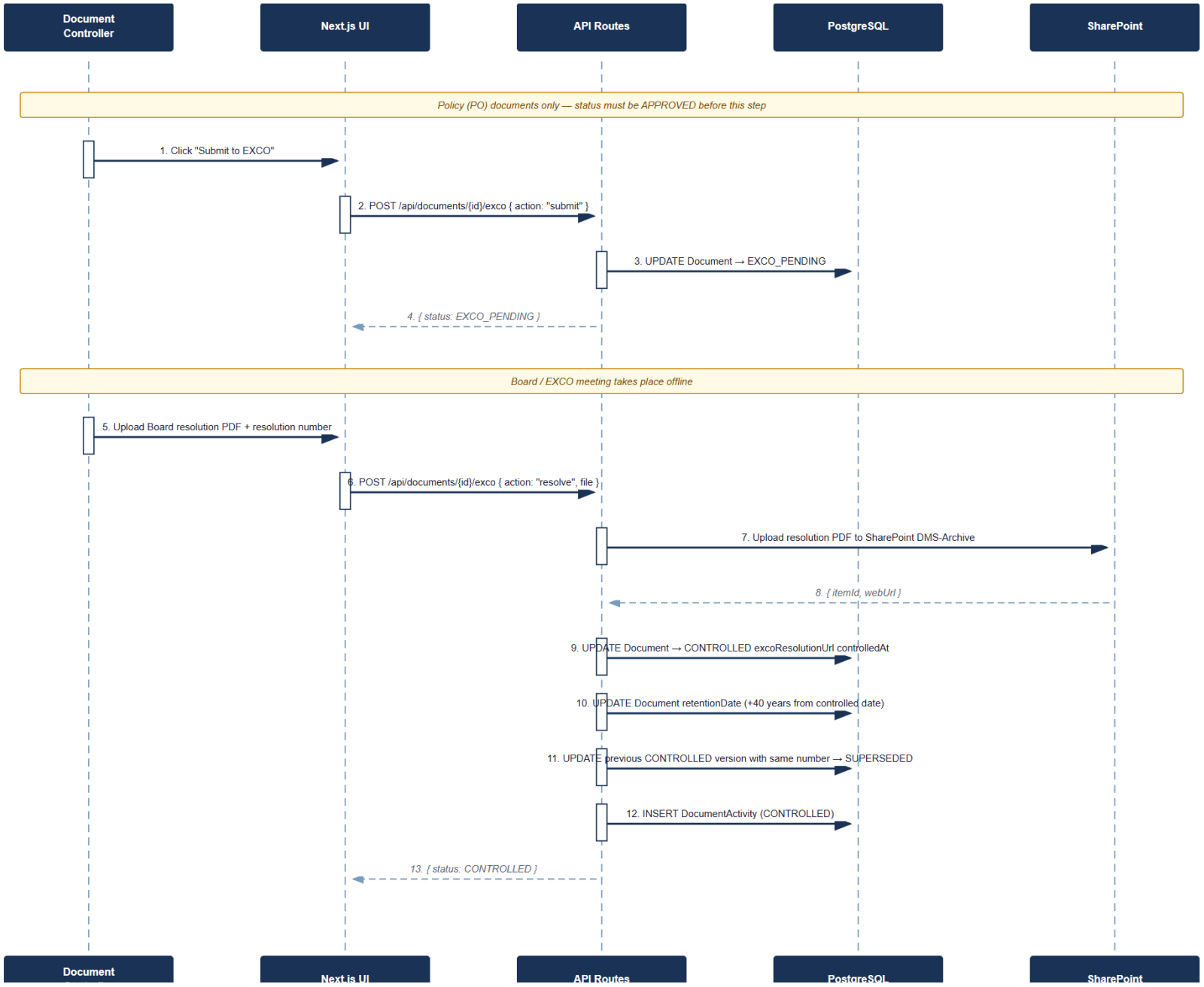

16. Changes requested by reviewer — notification sent
[Reviewer rejected]
17. UPDATE Document → REJECTED
18. UPDATE all other reviews → halted
19. Document rejected — reason provided
20. { success: true }
21. "Your decision has been recorded"
Reviewer
Email Token Link / Portal
API Routes
PostgreSQL
Power Automate (Email)
Document Controller

5. Sequence: Approval & Controlled Release



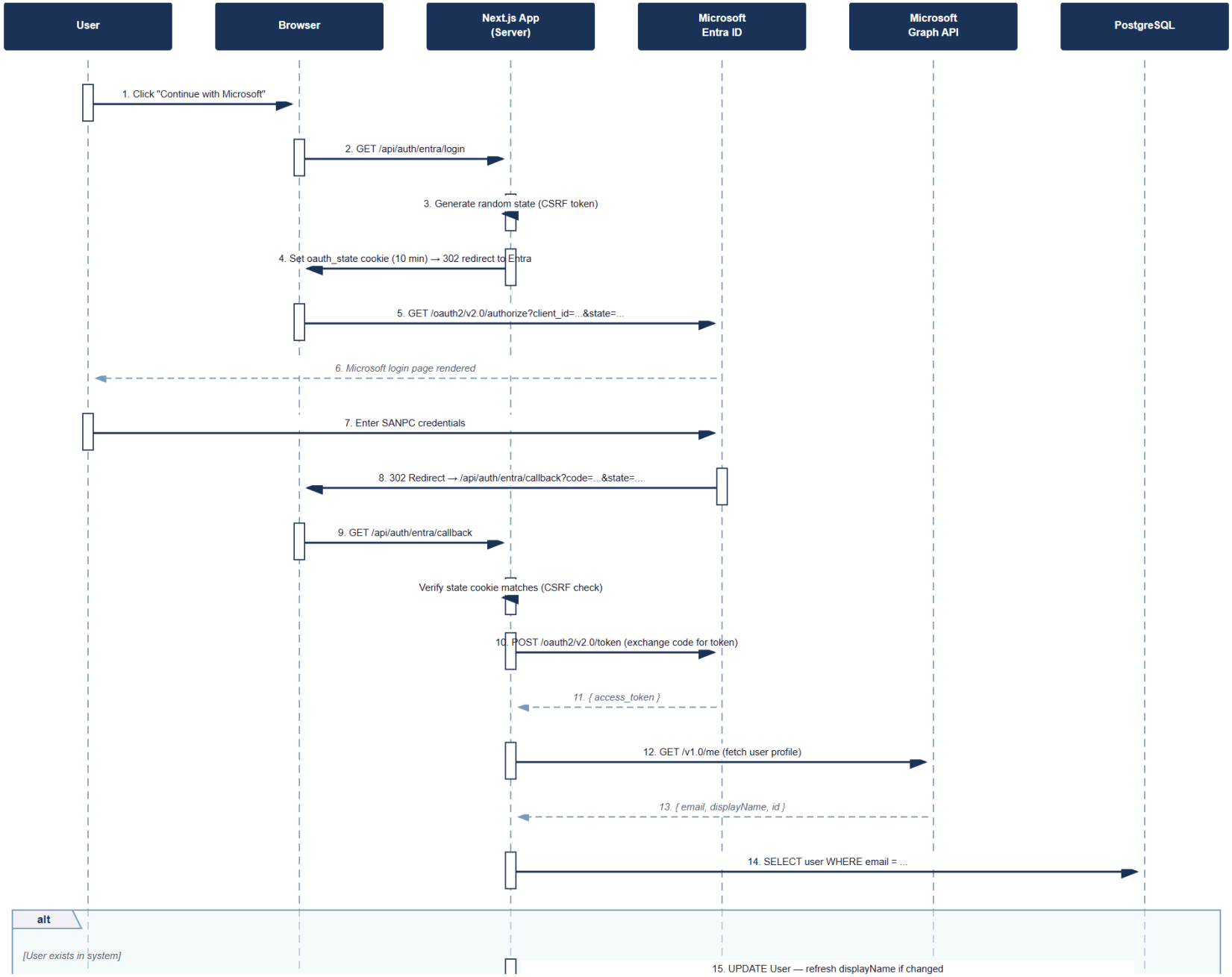

[An approver rejects]
17. UPDATE Document → REJECTED
18. Halt all other approvals
19. Document rejected — reason provided
Document Controller
Next.js UI
API Routes
PostgreSQL
SharePoint
Power Automate
Approver(s)

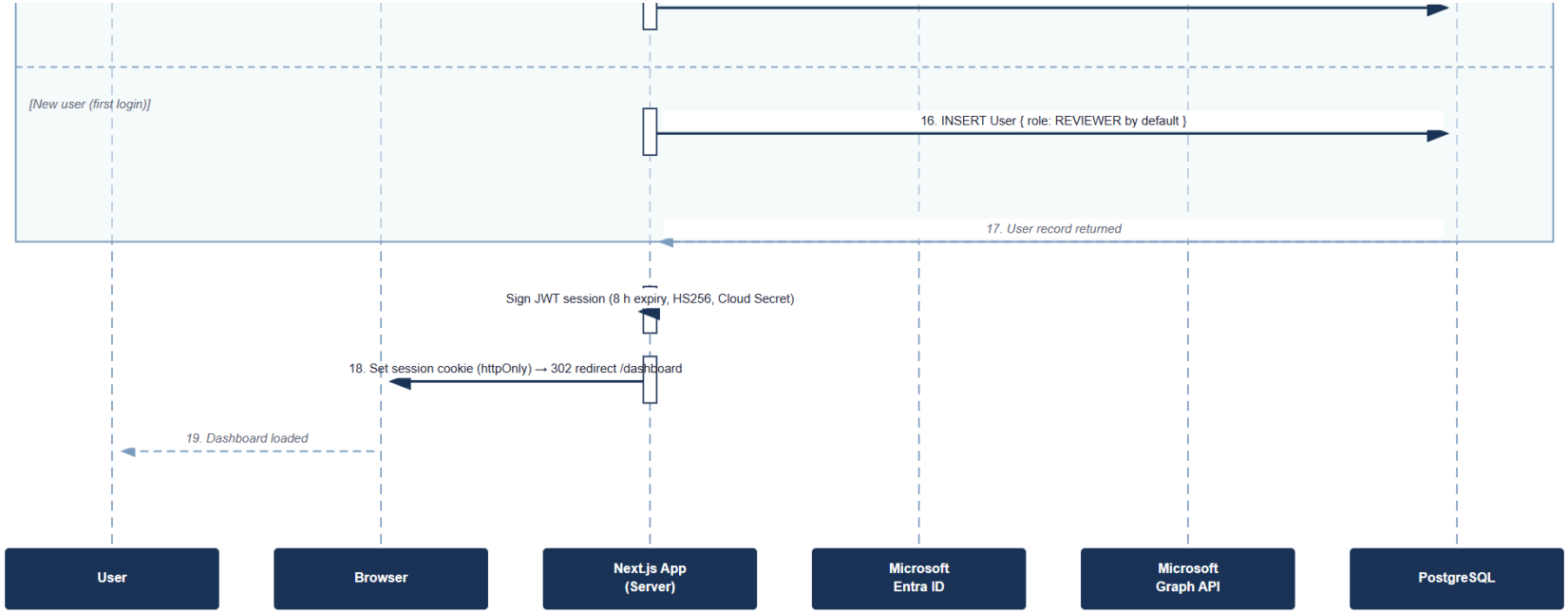
6. Sequence: Policy EXCO Workflow



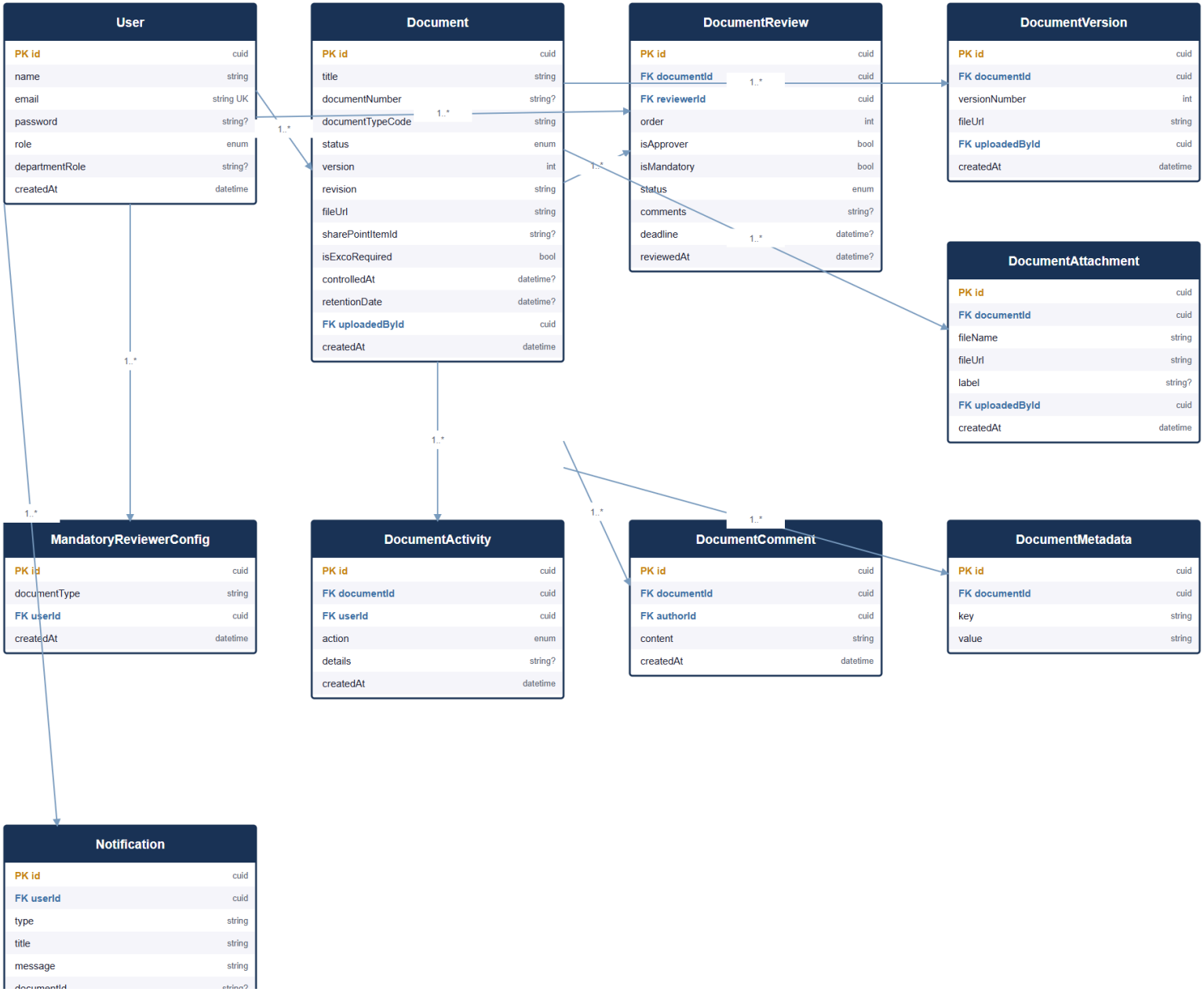

Controller

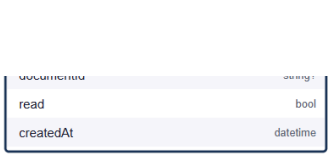
7. Sequence: SSO Authentication Flow




[New user (first login)]
16. INSERT User { role: REVIEWER by default }
17. User record returned
Sign JWT session (8 h expiry, HS256, Cloud Secret)
18. Set session cookie (httpOnly) → 302 redirect /dashboard
19. Dashboard loaded
User
Browser
Next.js App (Server)
Microsoft Entra ID
Microsoft Graph API
PostgreSQL

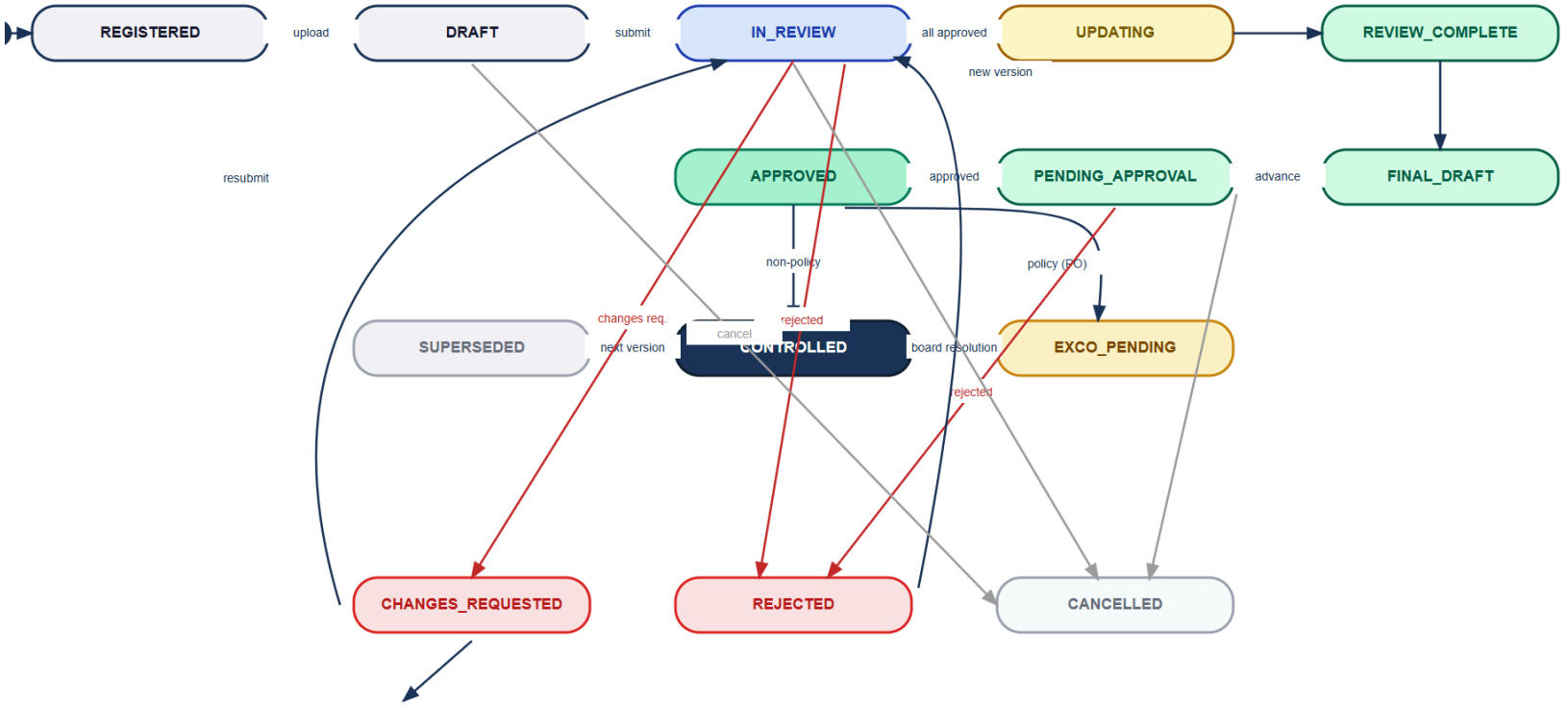
8. Entity Relationship Diagram





PK = Primary Key · FK = Foreign Key · UK = Unique · ? = nullable · Lines indicate one-to-many (1..*) relationships

9. Document Status Flow




Registered / Draft
Under Review
Review / Approval Complete
CONTROLLED
Rejected / Changes
Superseded / Cancelled
EXCO Pending